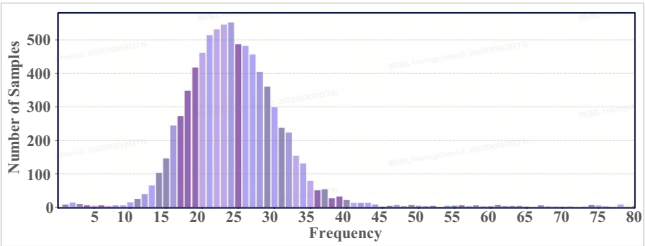

Figure 7: Distribution of keyword frequencies in the MISP-QEKS dataset.

A Forward-backward Algorithm

$$\mathrm{p}(q_t = i|\boldsymbol{o}_1 \cdots \boldsymbol{o}_{T_\mathrm{A}}, \lambda) = \alpha_t(i)\,\beta_t(i) / \sum_{i=1}^{N_\mathrm{S}} \alpha_t(i)\,\beta_t(i) \tag{13}$$

where $\alpha_t(i) = \mathrm{p}(\boldsymbol{o}_1 \cdots \boldsymbol{o}_t, q_t = i|\lambda)$ denotes the forward probability for i-th state at t-th time-step, andcan be computed as follow:

$$\alpha_1(i) = \pi_i\, b_i(\boldsymbol{o}_1), \quad \alpha_t(i) = \left[\sum_{j=1}^{N_\mathrm{S}} \alpha_{t-1}(j)\, a_{ji}\right] b_i(\mathbf{o}_t) \tag{14}$$

And $\beta_t(i) = \mathrm{p}(\boldsymbol{o}_{t+1} \cdots \boldsymbol{o}_{T_\mathrm{A}}|q_t = i, \lambda)$ is the backward probability of observing the remaining sequence for i-th state at t-th time-step, and can be computed as follow:

$$\beta_{T_\mathrm{A}}(i) = 1, \quad \beta_t(i) = \sum_{j=1}^{N_\mathrm{S}} a_{ij}\, b_j(\boldsymbol{o}_{t+1})\, \beta_{t+1}(j) \tag{15}$$

B Keyword Frequency Distribution

Figure 7 plots the frequency distribution of keywords in MISP-QEKS. The histogram exhibits a near-Gaussian profile and most keywords occur between 20 and 30 times. Such behaviour mirrors the spontaneous lexical usage patterns, indicating that MISP-QEKS faithfully preserves the real-world natural diversity.

C AUC and EER Computation

Given that

$$\begin{aligned} \mathrm{TPR}_i &= \frac{\mathrm{TP}_i}{\mathrm{TP}_i + \mathrm{FN}_i} \\ \mathrm{FPR}_i &= \frac{\mathrm{FP}_i}{\mathrm{FP}_i + \mathrm{TN}_i} \\ \mathrm{FNR}_i &= \frac{\mathrm{FN}_i}{\mathrm{TP}_i + \mathrm{FN}_i} \end{aligned} \tag{16}$$

where TP_i, FP_i, TN_i, FN_i are the counts of true positives, false positives, true negatives, and false negatives at threshold i, respectively. The AUC quantifies the overall ability of the classifier to distinguish between positive (keyword present) and negative (keyword absent) instances. In the context of discrete data, the AUC is computed by approximating the area under the ROC curve using the trapezoidal rule:

$$\mathrm{AUC} = \sum_{i=1}^{n-1} \frac{(\mathrm{FPR}_{i+1} - \mathrm{FPR}_i)(\mathrm{TPR}_{i+1} + \mathrm{TPR}_i)}{2} \tag{17}$$

The Equal Error Rate (EER) is defined as the point where the false positive rate (FPR) equals the false negative rate (FNR), indicating a balanced trade-off between the two types of errors. In practice, it is computed by interpolation between the closest points:

$$\mathrm{EER} = \frac{\mathrm{FPR}_i + \mathrm{FNR}_i}{2}, \quad i = \arg\min_i |\mathrm{FPR}_i - \mathrm{FNR}_i| \tag{18}$$

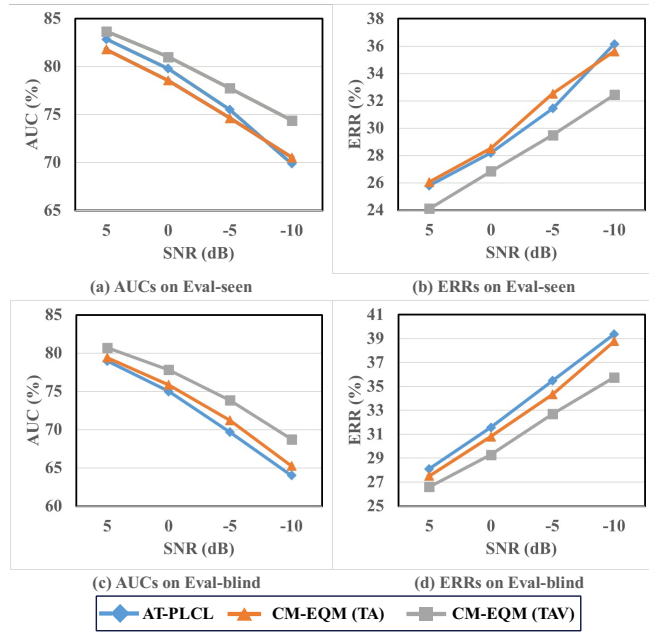

Figure 8: Line plots of average AUC and EER scores (in %) as a function of SNR levels (in dB) on the MISP-QEKS Eval-seen and Eval-blind splits.: (a) AUCs on Eval-seen, (b) EERs on Eval-seen, (c) AUCs on Eval-blind, and (d) EERs on Eval-blind.

D Performance Analysis of XEQ-Matcher Across SNR Levels

To isolate the contribution of the visual modality, we also plot average AUC and EER scores as a function of SNR on the MISP-QEKS Eval-seen and Eval-blind splits. Figure 8 (a) and (b) show that on Eval-seen, the XEQ-Matcher with visual input consistently outperforms its audio-only counterpart at every SNR: AUC is uniformly higher and EER uniformly lower. The gap widens as SNR decreases, indicating that visual cues supply complementary evidence when acoustic conditions deteriorate. A comparable trend appears on Eval-blind, as shown in Figure 8 (c) and (d): although the absolute scores drop because the keywords are unseen during training, the audio-visual model still retains a clear advantage across the entire SNR range, with the most significant margins again observed in the lowest-SNR regime. These results confirm that visual information systematically enhances robustness to noise for both in-vocabulary and out-of-vocabulary keywords.